

Assignment 2: Covariance, VaR, and Copulas

FinTech 545 - Quantitative Risk Management

Conventions

Returns are arithmetic, $r_t = P_t/P_{t-1} - 1$, expressed as decimals unless a percentage is shown. For dollar exposures v , portfolio P&L is $v^T r$. I use the **loss convention**

$$\text{VaR}_\alpha(P\&L) = -q_\alpha(P\&L), \quad \text{ES}_\alpha(P\&L) = -E[P\&L \mid P\&L \leq q_\alpha].$$

Thus positive numbers denote losses. A negative VaR denotes a gain at the quantile; I do not take an absolute value or floor it at zero. This distinction matters for the bonds in Problem 3. The default tail probability is 5%. Problems 2, 4, and 5 concern one day; Problem 3 concerns one year.

Historical and simulated risk follow `library/RiskStats.jl`. After sorting n P&Ls, the quantile is the average of the observations numbered $\lfloor n\alpha \rfloor$ and $\lceil n\alpha \rceil$, using one-based indices. Here $n\alpha$ is always an integer, so this is just that order statistic. ES averages all observations at or below it. There are no boundary ties in these calculations. Historical estimates use the supplied scenarios directly, without adding bootstrap sampling noise.

Descriptive variance and standard deviation use divisor $n - 1$. Skewness is $m_3/m_2^{3/2}$ and **excess** kurtosis is $m_4/m_2^2 - 3$, where $m_j = n^{-1} \sum_i (x_i - \bar{x})^j$. This matches the uncorrected standardized moment convention discussed in Week 1. Normal MLE scale instead uses divisor n . Student t scale is its location-scale parameter, not its standard deviation.

The predictions below were recorded after the allowed descriptive summaries and plots, before correlations, model fits, VaR, or ES were computed. They are retained even when the results disagree. `assignment2.py` computes all tables and figures; this report reads its numerical output. Each required simulation uses exactly 100000 draws with seed 545.

Problem 1: Correlations from Mismatched Histories

Predict

The joint observation counts are as follows. Diagonal entries are each series' available history. Only **81** rows contain all five series.

	IDX	A	B	C	D
IDX	250	242	242	228	96
A	242	242	242	221	93
B	242	242	242	221	93
C	228	221	221	228	83
D	96	93	93	83	96

(a) Which estimator is guaranteed PSD? Complete-case correlation is PSD. If Z contains centered, standardized complete observations, its correlation matrix is $Z^T Z / (n_c - 1)$. For every vector a , $a^T Z^T Z a / (n_c - 1) = \|Za\|^2 / (n_c - 1) \geq 0$. This is a common data matrix for every entry. Pairwise deletion uses different rows, means, and standard deviations for different pairs. Individually valid correlations can then be mutually inconsistent, with no single Gram matrix behind them. Pairwise correlation can therefore have a negative eigenvalue. PSD does not by itself guarantee strictly positive eigenvalues or an ordinary Cholesky factorization.

(b) Why is this data set exposed? With exact index replication, $r_{IDX} - .4r_A - .3r_B - .2r_C - .1r_D = 0$. The covariance matrix would have a nonzero null vector and a zero eigenvalue. Rescaling by nonzero standard deviations preserves rank, so correlation would also be singular. Small tracking error makes the smallest true eigenvalue small and positive rather than exactly zero. The matrix is therefore close to the PSD boundary: relatively small incompatible estimation errors can push this eigenvalue below zero. Strong index-stock relationships do not protect against that failure.

(c) Which estimates are least reliable? My prediction is that correlations involving D will be least reliable, especially C-D, with only 83 joint observations. D has a short listing history, and C's holidays reduce their overlap further. D's history also represents a different calendar window, so any time variation in dependence matters as well as sample size. A and B share 242 days and should be more stable. Complete-case estimation puts even these longer histories on the same short sample; its PSD guarantee comes at an information cost.

Fit

(d) Original correlation matrices, spectra, and Cholesky. The complete-case correlation matrix, using the common rows only, is:

	IDX	A	B	C	D
IDX	1.000000	0.871358	0.670860	0.564842	0.571930
A	0.871358	1.000000	0.396728	0.246622	0.413553
B	0.670860	0.396728	1.000000	0.316887	0.236053
C	0.564842	0.246622	0.316887	1.000000	0.185180
D	0.571930	0.413553	0.236053	0.185180	1.000000

The pairwise correlation matrix, before any repair, is:

	IDX	A	B	C	D
IDX	1.000000	0.869584	0.732867	0.713575	0.605691
A	0.869584	1.000000	0.468626	0.430888	0.438449
B	0.732867	0.468626	1.000000	0.463023	0.276692
C	0.713575	0.430888	0.463023	1.000000	0.184615
D	0.605691	0.438449	0.276692	0.184615	1.000000

Eigenvalues below are ordered from smallest to largest. Computations use the unrounded matrices; rounding a nearly singular matrix before factorization can change the result.

Eigenvalue	Complete case	Pairwise
1	1.505787e-03	-1.107185e-02
2	5.483453e-01	4.714220e-01
3	6.933196e-01	5.283668e-01
4	8.704410e-01	8.573811e-01
5	2.886388e+00	3.153902e+00

The complete-case minimum is 1.505787e-03; ordinary NumPy Cholesky **succeeds**. Its largest reconstruction error is 2.775558e-17. The pairwise minimum is -1.107185e-02; ordinary Cholesky **fails: the matrix is not positive definite**. This negative eigenvalue is far too large to dismiss as floating-point error.

(e) Full-history scaling and tracking variance. Let D_s contain each series' full-history sample standard deviation. The required covariance is $\Sigma = D_s C_{PW} D_s$, rather than a direct table of pairwise covariances. The standard deviations used are:

Series	Full-history standard deviation
IDX	0.01228141
A	0.01650183
B	0.01191893
C	0.01832355
D	0.02282866

In the order IDX, A, B, C, D,

$$w = (1, -0.4, -0.3, -0.2, -0.1)^T, \quad \text{Var}(w^T r) = w^T \Sigma w.$$

The calculated variance is **-1.799370e-06** in squared-return units. A negative variance is invalid, even though each individual standard deviation and pairwise correlation looks admissible.

(f) Two repairs. Rebonato-Jackel uses the spectral construction from Week 3: clip negative eigenvalues of $C_{PW} = Q\Lambda Q^T$ to zero, set $B = Q\sqrt{\Lambda_+}$, normalize each row of B to unit length, and return BB^T . This restores the diagonal but does not generally minimize distance.

Higham solves $\min_C \|C - C_{PW}\|_F$ subject to $C \succeq 0$ and $\text{diag}(C) = 1$, with $W = I$. I use the course's alternating PSD and unit-diagonal projections with the Dykstra correction. The squared-distance change and minimum-eigenvalue tolerance are both tightened to 10^{-12} . Convergence takes 33 iterations. In both repairs, the full-history standard deviations remain fixed.

Repair	Minimum eigenvalue	Frobenius distance	Tracking variance
Rebonato-Jackel	1.017081e-15	0.01618210	6.807298e-07
Higham	-8.618875e-13	0.01552000	6.787649e-07

Both matrices are PSD within numerical tolerance, and both tracking variances are positive. Higham's very small negative eigenvalue is its stopping tolerance, not the economically meaningful negative eigenvalue of the original matrix. These repairs lie near the PSD boundary; strict positive definiteness is not their objective.

Reconcile

(g) Why the negative variance proves failure. The definition of PSD is $a^T \Sigma a \geq 0$ for **every** vector a . The tracking weights provide an explicit counterexample. Equivalently, $w^T D_s C_{PW} D_s w$ is a negative quadratic form of the pairwise correlation matrix. There is no real-valued random portfolio with negative variance, so this is an inconsistency in the estimated dependence structure, not an unusually effective hedge. The complete-case PSD result and the failure of pairwise Cholesky match my prediction.

(h) Where Higham makes the changes. The following lists every distinct off-diagonal entry, sorted by absolute change. Its symmetric partner moves by the same amount.

Pair	Joint days	Original	Higham	Change
IDX-A	242	0.869584	0.862540	-0.007045
IDX-C	228	0.713575	0.708810	-0.004765
IDX-B	242	0.732867	0.728687	-0.004180
IDX-D	96	0.605691	0.601964	-0.003727
A-C	221	0.430888	0.433165	0.002277
A-B	242	0.468626	0.470623	0.001997
A-D	93	0.438449	0.440230	0.001781
B-C	221	0.463023	0.464374	0.001351
C-D	83	0.184615	0.185820	0.001205
B-D	93	0.276692	0.277749	0.001057

The largest adjustments are IDX-A, IDX-C, and IDX-B, rather than C-D. In part (c), I identified sampling reliability; Higham addresses global algebraic consistency. The index is almost spanned by the stock basket, so its high correlations load heavily on the nearly null direction. Reducing those entries helps remove the negative eigenvalue with relatively little total squared movement. The objective treats all entries with equal weight and knows neither their observation counts nor their sampling errors. A heavily sampled entry can therefore move more than a weakly sampled one. The repair restores feasibility; it does not certify which empirical estimate is statistically best.

(i) Estimator choice versus repair choice.

Comparison	Frobenius distance
Pairwise versus complete case	0.42393474
Rebonato-Jackel versus pairwise	0.01618210
Higham versus pairwise	0.01552000
Rebonato-Jackel versus Higham	0.00457878

The estimator gap is about 27.3 times the Higham repair distance. **The choice of estimator matters much more for these data than the choice between the two repairs.** Higham improves the distance modestly, and the two repaired tracking variances are almost identical. The larger substantive choice is whether to discard most of the longer histories to force a common sample. This does not make repair optional: the unrepaired pairwise matrix remains invalid regardless of how small its repair is.

Problem 2: A Volatility Estimate After the Regime Changed

Predict

I compute 500 returns and subtract their sample mean, 0.00016188. The plot shows a visibly wider range of moves near the end. Before fitting, I choose the last 40 observations, days 461-500, as an approximate new regime. This is a visual boundary, not an estimated change point.

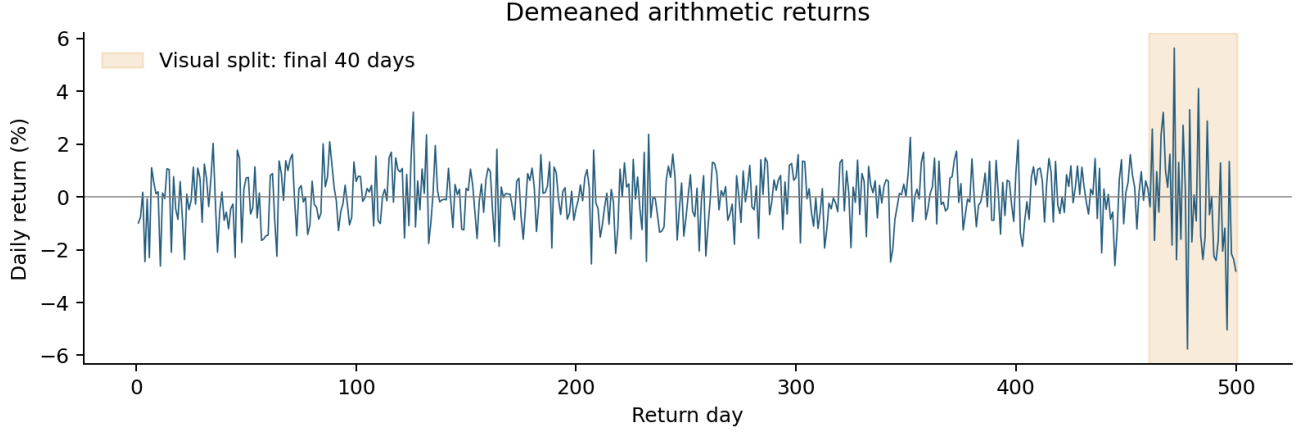


Figure 1: The shaded interval marks the regime split chosen from the plot.

(a) **Predicted ordering, smallest to largest.** My original prediction is:

Historical < t MLE < Normal, equal weight < EW Normal (.97) < EW Normal (.94).

Historical simulation gives most weight to the older quiet period. I expect a t fit to keep a relatively narrow center while accommodating a few extremes, whereas an equal-weight normal spreads their contribution to variance across its entire distribution. EW estimates should be larger because they emphasize the recent turbulent period; .94 should respond more strongly than .97. The historical-versus- t ordering is the least certain part of this prediction: a 5% quantile is not determined by kurtosis alone.

(b) **Memory of the EW estimator.** Following `library/ewCov.jl`, observation $t = 1, \dots, n$ receives

$$\hat{w}_t = \frac{(1 - \lambda)\lambda^{n-t}}{1 - \lambda^n}, \quad \bar{x}_w = \sum_t \hat{w}_t x_t, \quad \hat{\sigma}_w^2 = \sum_t \hat{w}_t (x_t - \bar{x}_w)^2.$$

The newest observation has the largest weight. Although the full sample is demeaned first, its EW mean is generally nonzero; the course covariance routine removes that EW mean as well. No additional unbiasedness correction is applied. The quantities requested are

$$n_{\text{eff}} = \frac{1}{\sum_t \hat{w}_t^2} = \frac{1 + \lambda}{1 - \lambda} \frac{1 - \lambda^n}{1 + \lambda^n}, \quad h = \frac{\log(0.5)}{\log \lambda}, \quad W_{\text{recent}} = \frac{1 - \lambda^{40}}{1 - \lambda^{500}}.$$

Lambda	Finite n_eff	Infinite n_eff	Half-life (days)	Recent weight
0.94	32.3333	32.3333	11.20	91.58%
0.97	65.6666	65.6667	22.76	70.43%

The final 40 days receive 91.58% of the .94 weight and 70.43% of the .97 weight, compared with 8.00% under equal weighting. The .94 estimate is consequently much closer to a measure of the recent regime than to a summary of all 500 days.

(c) First four moments and the single-regime prediction.

Series	Mean	Variance	Skewness	Excess kurtosis
Original returns	0.000162	1.418981e-04	-0.1556	2.1324
Demeaned returns	0.000000	1.418981e-04	-0.1556	2.1324

Demeaning changes only the first moment. Excess kurtosis of 2.1324 is well above the normal value of zero, while skewness is comparatively modest. I would not regard one homoskedastic normal regime as a reasonable explanation of the sample. A normal sample can have nonzero sample kurtosis by chance, but the time plot provides an additional explanation: periods with different variances are being pooled.

Fit

(d) Five one-day VaRs. Normal VaR uses $10^6 z_{0.95} \hat{\sigma}$ with zero mean after demeaning. Student t MLE estimates location as well as scale and degrees of freedom on these same demeaned observations, so its small fitted location is retained in its quantile:

$$\text{VaR}_t = -10^6 \left(\hat{\mu}_t + \hat{s}_t t_{\hat{\nu}}^{-1}(0.05) \right).$$

Method	One-day 5% VaR (\$)
Normal, equal weight	\$19,594
Normal, EW 0.97	\$34,064
Normal, EW 0.94	\$36,308
Student t MLE	\$18,872
Historical	\$20,574

t parameter	MLE
Degrees of freedom	8.269941
Location	0.00018162
Scale	0.01028999

Volatility estimator	Daily standard deviation	EW mean removed
Equal weight	1.1912%	Sample mean already removed
EW 0.97	2.0709%	-0.00375469
EW 0.94	2.2074%	-0.00815646

All dollar figures use a fixed \$1,000,000 exposure. The larger negative EW sample means are removed for the variance calculation, not used as forecasts of tomorrow's drift. Including them as expected returns would answer a different question from comparing volatility estimates on demeaned returns.

Reconcile

(e) **What happened to the predicted order?** The observed order is

$$t \text{ MLE} < \text{Normal, equal weight} < \text{Historical} < \text{EW Normal } (.97) < \text{EW Normal } (.94).$$

I correctly predicted the EW ordering and that both EW values would exceed the full-history estimates. I incorrectly put historical VaR below both the t and equal-weight normal. The historical 5% threshold is set by the actual lower-tail observations, not just by the proportion of calm days. Here it is more negative than either fitted full-history 5% quantile. The t fit can assign a relatively small scale to the center and use its heavier far tails for the largest moves. Therefore it need not have a larger 5% VaR than a normal fitted to the overall variance. My original reasoning about the center did not reliably locate the empirical 5% order statistic.

(f) **Regime volatilities and excess kurtosis.** Keeping the visual split fixed:

Return days	Observations	Standard deviation	Excess kurtosis
1-460	460	1.0178%	-0.2131
461-500	40	2.4289%	-0.1199

The recent standard deviation is about 2.39 times the earlier one. For two conditionally normal, common-mean regimes with probabilities p_j and variances s_j^2 , the pooled excess kurtosis is

$$\kappa_{\text{excess}} \approx 3 \frac{\sum_j p_j s_j^4}{(\sum_j p_j s_j^2)^2} - 3.$$

Using the observed regime variances gives 2.5723, compared with the actual pooled value 2.1324. This is an illustration, not an exact identity for the data: the regime sample means differ, within-regime samples are finite, and the common-mean normal approximation is imperfect. Nevertheless, the large variance contrast explains why pooled returns have a sharp center and heavy tails even though each regime's sample excess kurtosis is near zero. The fitted t is describing this unconditional mixture of quiet and turbulent days. It has no time index and does not identify the current regime or predict when it will change.

(g) **Responsiveness versus estimation noise.** Week 3 gives $SE(\hat{\sigma}) \approx \hat{\sigma}/\sqrt{2n_{\text{eff}}}$. For a zero-mean normal VaR, multiplying by $10^6 z_{0.95}$ gives the corresponding dollar SE.

Lambda	SE of daily volatility	Approximate SE of VaR (\$)
0.94	0.2745%	\$4,515
0.97	0.1807%	\$2,972

The EW VaR gap is \$2,244, smaller than either individual approximate SE. An independence benchmark for the SE of their difference would be \$5,406. The estimators use the same observations and are positively related, so that benchmark is not the actual SE of the difference; these numbers are not a formal significance test. The course approximation, itself based on a stable volatility setting, does not support describing the observed gap as a clearly large separation from estimator noise, particularly across a regime change.

My opinion: I would use .94 for a trading limit because it responds sooner to the recent increase, accepting the noisier limit. I would use .97 for a capital number because its larger effective sample size produces a steadier estimate. I would still accompany a capital measure with stress analysis: choosing a slower decay does not make a stale volatility forecast safe after a persistent regime shift.

Problem 3: When Diversification Raises VaR

Predict

The supplied rows are joint one-year return scenarios. I preserve their pairing when forming the diversified P&L. The plots use a logarithmic frequency axis so that the default region is visible alongside the large performing-bond peak.

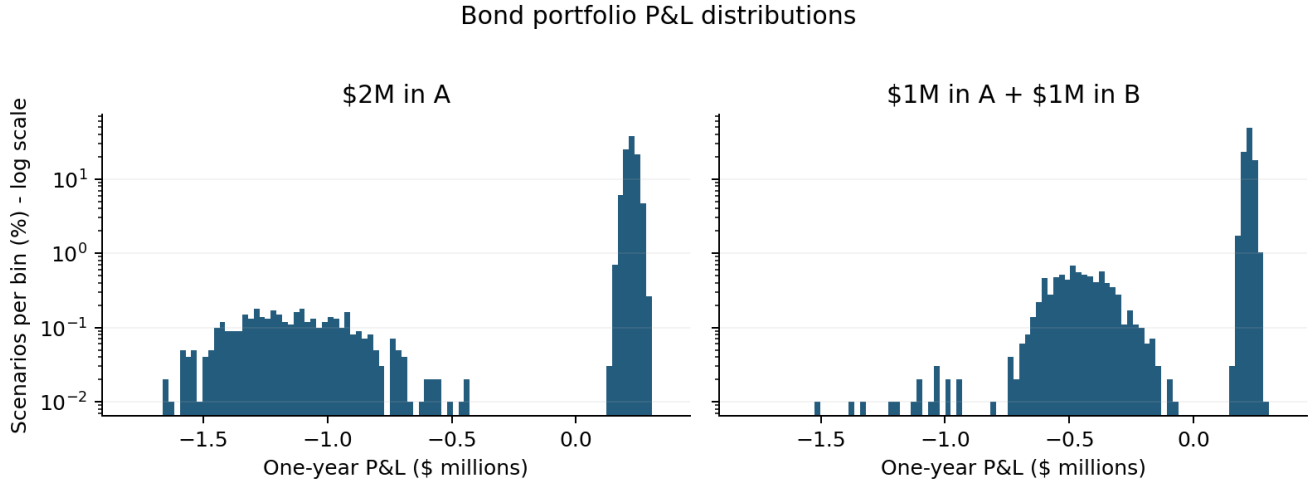


Figure 2: Both investment choices use the same total exposure.

(a) Marginal moments and the normal approximation.

Series	Mean	Variance	Skewness	Excess kurtosis
A	0.083709	1.849265e-02	-4.8225	22.0820
B	0.085011	1.745088e-02	-4.9148	23.0744

Each bond has strongly negative skewness and very large excess kurtosis. Most scenarios earn approximately the purchase-to-payoff gain, while a small group has a severe default loss. A single normal distributes probability smoothly around one center; it cannot represent this separation between survival and default outcomes. Its standard deviation is influenced by defaults even when the chosen quantile is still in the performing-bond region. That is why normal VaR may give a particularly misleading answer at 5%.

(b) Counts and prediction before risk calculations. A loss above 20% means $r < -.20$.

Event	Scenarios	Fraction
A loss >20%	406	4.06%
B loss >20%	393	3.93%
At least one loss >20%	782	7.82%
Both losses >20%	17	0.17%

My prediction is **higher 5% VaR but lower 5% ES after diversification**. For either individual bond the severe-loss frequency is below 5%, so its 5% quantile can sit outside the default group. For the split investment, at least one bond has a severe loss in more than 5% of scenarios, moving that quantile into the one-default region. ES should fall because losing on one half of the investment is much less severe

than losing on the entire investment, and simultaneous defaults are rare. The plot supports that severity comparison; the counts alone do not determine loss magnitudes.

Fit

(c) Historical 5% VaR and ES.

Position	Historical 5% VaR (\$)	Historical 5% ES (\$)
1M A	-\$84,531	\$443,628
1M B	-\$86,011	\$419,587
2M A	-\$169,061	\$887,257
1M A + 1M B	\$414,767	\$541,997

The negative single-bond VaRs are intentional. For example, the 5% P&L quantile of the concentrated \$2M position is a **gain** of \$169,061; its loss-convention VaR is the negative of that gain. Yet its worst 5% average is a large loss. Taking absolute values would erase precisely the quantile behavior this problem is designed to examine.

(d) Normal VaR from the P&L moments. Here I retain the sample one-year mean and calculate $-\bar{P} + z_{0.95} s_P$. Demeaning would discard the expected bond payoff and change the requested risk measure.

Position	Mean P&L (\$)	P&L SD (\$)	Normal 5% VaR (\$)
1M A	\$83,709	\$135,988	\$139,971
1M B	\$85,011	\$132,102	\$132,277
2M A	\$167,418	\$271,975	\$279,941
1M A + 1M B	\$168,721	\$189,796	\$143,466

(e) Historical results at 1%.

Position	Historical 1% VaR (\$)	Historical 1% ES (\$)
1M A	\$650,545	\$709,548
1M B	\$641,248	\$700,805
2M A	\$1,301,090	\$1,419,096
1M A + 1M B	\$598,237	\$716,408

The concentrated 1% quantile now lies well within its default outcomes. Splitting the investment lowers both its VaR and ES at this threshold.

Reconcile

(f) Explicit subadditivity checks at 5%. Here A and B in the inequalities mean the separate \$1M positions, with A+B their combined \$2M position.

Measure	Risk(A+B) (\$)	Risk(A)+Risk(B) (\$)	Subadditive?
VaR	\$414,767	-\$170,542	No
ES	\$541,997	\$863,216	Yes

For VaR, the right-hand side is the sum -\$84,531 plus -\$86,011, giving -\$170,542. The combined VaR of \$414,767 is larger, so the inequality fails. For ES, the standalone values are \$443,628 and \$419,587; their sum exceeds the combined \$541,997. Thus ES passes the same check.

The mechanism matches the prediction. There are 500 observations in a 5% tail. A has only 406 severe-loss observations and B has only 393, so a standalone 5% quantile reaches the performing-bond group. Combining the bonds creates 782 scenarios with at least one severe loss, so the combined quantile reaches a default outcome instead. VaR compares these thresholds and ignores how bad the observations beyond them are. ES includes their severity; the mostly different default scenarios then provide a diversification benefit rather than a discontinuous quantile penalty.

(g) What normal VaR gets right and wrong. The normal calculation favors the diversified portfolio, at \$143,466, over the concentrated portfolio, at \$279,941. The split portfolio has nearly the same expected gain but a much smaller standard deviation, so a normal location-scale model must favor it. This is **wrong for the actual 5% VaR comparison**, which goes in the other direction. It happens to agree with the ES-based preference for diversification, but for an incomplete reason: a lower second moment does not describe where a rare default cluster sits relative to the 5% quantile. Matching mean and variance cannot recover that feature of a mixture distribution.

(h) Why the 1% answer changes. At 1%, even a single bond's default frequency is greater than the tail probability. Both choices' quantiles are now inside loss regions, so the quantile no longer skips the concentrated default risk. The concentrated VaR becomes \$1,301,090, versus \$598,237 for the split portfolio. The empirical both-default frequency is only 0.17%, below 1%, and many diversified tail scenarios still involve just one default. The tail exposure is therefore usually only half the investment. VaR's subadditivity failure depends on where the selected probability falls relative to the event frequencies; it need not appear at every confidence level for the same positions.

Problem 4: Gaussian or t Copula

Predict

(a) Marginal moments and a single-observation check.

Series	Mean	Variance	Skewness	Excess kurtosis
X1	0.000257	1.414697e-04	-0.0342	-0.1225
X2	-0.000252	2.614199e-04	-0.6141	7.0834
X3	0.000890	1.003768e-04	0.0740	2.8537

From moments alone, I predict a normal margin for X1 and t margins for X2 and X3. X1 has little skewness or excess kurtosis. X2 and X3 have visibly heavier tails. X2's negative skewness and unusually large kurtosis deserve a sensitivity check because fourth powers can give one observation enormous influence.

The largest absolute centered return in each series happens to be observation 952 (one-based row number, excluding the CSV header). The following deletes that one observation **for diagnosis only**.

Series	Return deleted	Skew, full	Skew, without	Excess, full	Excess, without
X1	-4.0000%	-0.034	-0.005	-0.123	-0.193
X2	-14.2914%	-0.614	0.056	7.083	1.710
X3	-6.1244%	0.074	0.313	2.854	1.716

For X2, removing this observation changes excess kurtosis from 7.0834 to 1.7101 and skewness from -0.6141 to 0.0558. The evidence for heavy tails remains, but the apparent asymmetry largely disappears. I would not infer a stable asymmetric distribution from that one large loss. A symmetric t is a reasonable candidate, with the limitation that it cannot model skewness. **All observations are retained in every model fit and historical risk calculation.**

(b) Rank uniforms, joint counts, and the copula prediction. I use $U_{ij}^{rank} = \text{rank}(x_{ij}) / (n + 1)$, with average ranks for ties. The course's fitted-copula routine uses fitted CDFs rather than prescribing a separate rank plotting position; this rank convention keeps the preliminary plot strictly inside $(0, 1)$ as requested. Its observed range is 0.000999 to 0.999001.

Rank uniforms: all three asset pairs

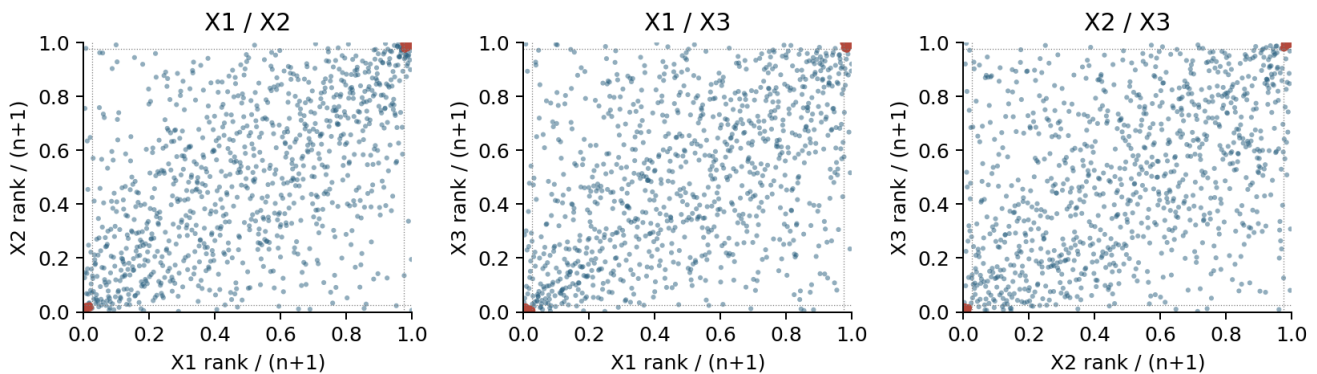


Figure 3: Every pair of rank uniforms. Red points are joint lower or joint upper 2.5% observations.

For each pair, a lower joint-tail day has both ranks at or below .025; an upper joint-tail day has both at or above .975. Each individual tail contains 25 days. Under independence, the expected overlap in either direction is $n(.025)^2 = 0.625$ days.

Pair	Both worst 2.5%	Both best 2.5%	Independence, each tail
X1-X2	6	8	0.625
X1-X3	7	12	0.625
X2-X3	6	9	0.625

The plots have broad central clouds but crowded same-direction corners. The counts are much larger than independence predicts, so I expect the ***t* copula to win**. Positive correlation in a Gaussian copula also increases finite-tail counts; exceeding independence alone is not proof of *t* tail dependence. My prediction comes from the corner concentration relative to the broad center, and must be checked against both fitted models.

Fit

(c) Marginal maximum likelihood and AICc. For a normal margin there are $k = 2$ fitted parameters, location and scale. For a *t* margin there are $k = 3$, location, scale, and degrees of freedom. I use

$$\text{AICc} = -2\ell + 2k + \frac{2k(k+1)}{n-k-1}.$$

Asset	Model	Location	Scale	nu
X1	Normal	0.0002566	0.0118882	–
X1	t	0.0002566	0.0118882	5.346806e+12
X2	Normal	-0.0002524	0.0161604	–
X2	t	-0.0001759	0.0125440	5.2715
X3	Normal	0.0008898	0.0100138	–
X3	t	0.0007456	0.0081807	6.0948

Asset	Model	k	Log likelihood	AICc
X1	Normal	2	3013.274	-6022.537
X1	t	3	3013.274	-6020.524
X2	Normal	2	2706.253	-5408.494
X2	t	3	2762.104	-5518.185
X3	Normal	2	3184.852	-6365.691
X3	t	3	3217.020	-6428.016

The selected margins are **X1 normal, X2 *t*, X3 *t***, agreeing with my moment-based predictions. For X1, numerical *t* fitting drives degrees of freedom to a very large value: this is the normal limit, not a meaningfully identified finite tail parameter. Its likelihood is effectively identical to the normal's, so the extra-parameter penalty selects the normal. The finite *t* fits for X2 and X3 gain enough likelihood to overcome that penalty. The reported *t* fits all satisfy the course's $\nu > 2$ condition.

(d) Fitted uniforms, correlation, and copula likelihoods. I now transform each observation using its **selected fitted marginal CDF**, $U_{ij} = \hat{F}_j(x_{ij})$. These are different from the ranks in (b). For

numerical safety, CDF outputs are bounded to $[\epsilon, 1 - \epsilon]$, with double-precision machine epsilon. In this data set 0 values require clipping; the fitted uniform range is 3.278162e-05 to 0.999675723.

The course comparison uses Kendall's τ and $R_{ij} = \sin(\pi\tau_{ij}/2)$, with unit diagonal. The matrices are:

Kendall's tau:

	X1	X2	X3
X1	1.000000	0.396448	0.335431
X2	0.396448	1.000000	0.307447
X3	0.335431	0.307447	1.000000

Copula correlation R:

	X1	X2	X3
X1	1.000000	0.583263	0.502851
X2	0.583263	1.000000	0.464383
X3	0.502851	0.464383	1.000000

The eigenvalues of R are 0.412792, 0.551919, 2.035289. The minimum, 0.412792, is positive. No correlation repair or positive-definite adjustment is needed.

For each observation the Gaussian and t copula log densities are calculated explicitly as

$$\ell_{G,i} = -\frac{1}{2} \log |R| - \frac{1}{2} z_i^T (R^{-1} - I) z_i, \quad z_{ij} = \Phi^{-1}(U_{ij}),$$

$$\ell_{t,i} = \log f_{t_d, \nu, R}(y_i) - \sum_j \log f_{t_\nu}(y_{ij}), \quad y_{ij} = t_\nu^{-1}(U_{ij}).$$

Here R is the multivariate t **scale** matrix inside the density; it is not multiplied by $(\nu - 2)/\nu$. Its diagonal is one so each transformed coordinate is a standard univariate t_ν . Confusing scale and covariance would give the wrong copula density.

I follow `Week05/choosing_between_copulas.jl` and `library/multivariate_t.jl` exactly for fitting: hold margins and R fixed; evaluate 200 equally spaced values of $\theta = 1/\nu$ on $[\cdot 01, \cdot 49]$; then refine with another 200 points between the neighbors of the best point. The actual search interval is $\nu \in [1/\cdot 49, 100]$. This is a profile fit, not a joint refit of all marginal and dependence parameters.

Copula	k	Fitted nu	Log likelihood	AICc	BIC
Gaussian	0	—	350.4403	-700.8805	-700.8805
Student t	1	3.659762	451.9995	-901.9950	-897.0912

Parameter count: as specified in the Week 5 comparison, $k_G = 0$ and $k_t = 1$. The margins and rank-estimated R have already been fixed for both models, so these are **conditional copula** criteria, not whole-joint-model criteria. I do not count R or the marginal parameters again. BIC is $-2\ell + k \log n$.

(e) Portfolio simulation and risk. Each copula produces exactly 100000 simulated days. With $LL^T = R$, the Gaussian latent vector is $Z = L\eta$ for independent standard normals. For the t copula I use $Y = Z/\sqrt{W/\hat{\nu}}$, where one independent $W \sim \chi_{\hat{\nu}}^2$ is shared by all three assets on a given day. Gaussian

uniforms are $\Phi(Z)$; t uniforms are $t_{\hat{\nu}}(Y)$. Each column then passes through its own selected marginal inverse CDF.

The normal vectors are shared between the two simulations (seed 545), while the chi-square draws use seed 546. This helps comparison without changing either fitted distribution. The copula degrees of freedom and each margin's degrees of freedom have distinct roles. Portfolio P&L is $10^6(X_1 + X_2 + X_3)$; fitted marginal locations are retained because this problem does not assume zero means.

Method	5% VaR (\$)	5% ES (\$)	1% VaR (\$)	1% ES (\$)
Gaussian	\$49,411	\$66,383	\$76,247	\$93,827
Student t	\$48,865	\$67,684	\$78,339	\$99,579
Historical	\$47,939	\$65,699	\$70,092	\$102,424

The historical row uses all 1000 observed joint returns. Its 1% ES averages only 10 days, whereas each simulated 1% ES averages 1000 draws. The historical tail is consequently sensitive to individual observations; simulation reduces Monte Carlo error but does not remove uncertainty about the fitted model.

(f) Simulated joint tails, expressed per 1,000 days. I re-rank each simulated asset and use the same own-worst/best 2.5% definition as in (b). The simulated count is multiplied by 1000/100000. Thus the entries below are expected counts over 1,000 days implied by the simulation, not integer counts from a second 1,000-day draw.

Pair	Observed low	Observed high	Gauss low	Gauss high	t low	t high
X1-X2	6	8	6.35	6.25	9.18	9.42
X1-X3	7	12	4.67	4.53	7.71	8.08
X2-X3	6	9	4.53	4.35	7.42	7.58

Reconcile

(g) Which copula wins, and does it reproduce the tail counts? The t copula gains 101.5592 in log likelihood. Its AICc is lower by 201.1144 and BIC is lower by 196.2107. Both criteria strongly favor t , agreeing with my prediction.

The t increases the implied same-direction tail counts for every pair. This is especially helpful for the X1-X3 and X2-X3 upper tails, where the Gaussian counts are well below the observed counts. It is not uniformly closer: for X1-X2's lower tail, the Gaussian is closer to the observed count of 6. The observed upper counts also exceed lower counts for all pairs, whereas both copulas are symmetric in their population upper and lower tails. A fitted t provides stronger joint tails but does not reproduce every finite-sample asymmetry. Its information-criterion win scores the full sample, not a minimum-error fit to this small table.

(h) Which risk number moves most?

Measure	t minus Gaussian (\$)	Percentage change
5% VaR	-\$546	-1.10%
5% ES	\$1,301	1.96%
1% VaR	\$2,093	2.74%
1% ES	\$5,752	6.13%

The **1% ES** moves most, increasing by \$5,752 or 6.13%. The 5% VaR changes by only -\$546. Both copulas have the same margins and the same rank-based correlation. The t 's shared scale shock mainly changes the probability of simultaneous extreme returns. A 5% quantile is a moderate threshold and does not average the severity of more distant observations. In contrast, 1% ES averages exactly those extreme joint losses. Heavier joint tails need not raise every quantile: redistributing mass toward both the center and the far tails can leave the 5% threshold nearly unchanged or slightly lower, as it is in this simulation.

(i) **Where the likelihood improvement comes from.** For every observation I calculate $d_i = \ell_{t,i} - \ell_{G,i}$ and save the individual values in `problem4_daily_loglik.csv`. I define a non-extreme day as one on which **all three empirical rank uniforms lie strictly between .05 and .95**. Thus no asset is in either of its own outer 5% tails; this is not a cutoff on total portfolio P&L.

Group	Days	Sum of t minus Gaussian log likelihood	Share of total
All observations	1000	101.5592	100.00%
Non-extreme (empirical ranks)	784	49.8017	49.04%
At least one extreme rank	216	51.7575	50.96%

About 49.04% of the gain comes from non-extreme days. A t copula changes the density throughout the cube, including the concentration near its center, not only the far corners. AICc and BIC sum log densities over **every** observation and then impose a parameter penalty. They select better overall probabilistic fit, not the model with the best fit to a chosen tail risk statistic. This explains how a decisive criterion gap can coexist with a small difference in 5% VaR.

As a definition sensitivity check, using the **fitted marginal uniforms** in (.05,.95) rather than the empirical own-tail ranks gives 793 non-extreme days, a contribution of 50.4794, and 49.70% of the improvement. The interpretation is the same under either precise tail classification.

(j) **Asymptotic tail dependence.** The most correlated pair is X1-X2, with $\rho = 0.583263$ and fitted copula $\nu = 3.659762$. Week 5 gives

$$\lambda_L = \lambda_U = 2t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right).$$

The resulting coefficient is **0.321952**. It is a limiting conditional probability: as one margin becomes arbitrarily extreme in one direction, the chance the other is equally extreme approaches this positive value.

The Gaussian copula coefficient is **zero** for $|\rho| < 1$. Its joint extreme probability decays faster than its marginal extreme probability, so their ratio, $P(Z_2 > t \mid Z_1 > t)$, tends to zero as $t \rightarrow \infty$. Symmetry gives the same lower-tail result. Imperfectly correlated Gaussian shocks can still produce substantial overlap at a finite threshold: zero asymptotic dependence does not mean independence or zero finite-2.5% overlap. The Gaussian counts in (f) demonstrate that distinction.

Problem 5: Model Based Simulation and Residual Correlation

Predict

(a) **Direction of the independent-residual shortcut.** Before fitting, I expect omitted common industry exposure to produce positive residual covariance. For exposures v_A, v_B , the residual contribution to portfolio variance is

$$v_A^2 \sigma_{\epsilon_A}^2 + v_B^2 \sigma_{\epsilon_B}^2 + 2v_A v_B \text{Cov}(\epsilon_A, \epsilon_B).$$

In P1 both exposures are positive, so the covariance term is positive. Setting it to zero removes real common risk and **understates P1 VaR**. In P2 the exposures have opposite signs, so the same positive residual covariance enters with a negative sign. It makes the spread less volatile because the residual moves offset; deleting it **overstates P2 VaR**. This argument assumes positive residual covariance and the stipulated zero-mean normal model. I also expect P2 to be more sensitive in percentage terms if similar market betas cancel, leaving residual dependence as a large part of its remaining risk.

Fit

(b) **Arithmetic returns and OLS.** I calculate returns on MKT, A, and B and fit both stocks with the same intercept-and-market design matrix:

$$r_A = \alpha_A + \beta_A r_{MKT} + \epsilon_A, \quad r_B = \alpha_B + \beta_B r_{MKT} + \epsilon_B.$$

Stock	Alpha	Beta	Residual SD (n-1)	Regression error SD (n-2)
A	0.00020501	1.035278	0.01288830	0.01290123
B	0.00031477	0.935417	0.01153844	0.01155002

There are 500 returns. Residual correlation is **0.698307**, supporting the predicted positive omitted dependence. The market's sample daily standard deviation is 1.0553%.

For simulation I use the sample covariance of the residual series, with divisor $n - 1$, matching the sample covariance convention in part (d). This makes the OLS covariance decomposition exact in sample. For clarity, the table also reports the usual regression error estimate using $n - 2$ after estimating an intercept and a slope; that slightly larger estimate is not used in the simulation. The residual covariance matrix in squared-return units is:

	A	B
A	0.000166108	0.000103846
B	0.000103846	0.000133136

(c) **Two simulations with only residual covariance changed.** I simulate 100000 days of a zero-mean normal market and normal residuals, independent of the market. For the full model, residual normals are multiplied by the Cholesky root of the matrix above. For the shortcut, only its off-diagonal entries are set to zero. Both runs use the same market draws, standard-normal residual inputs, fitted betas, and residual marginal volatilities.

The assignment assumes **zero expected returns**. I therefore report the estimated intercepts for the regression but simulate centered stock returns $r_i^* = \hat{\beta}_i r_{MKT}^* + \epsilon_i^*$. Equivalently, start with the fitted

intercept model and subtract each stock's fitted mean. Adding the fitted intercepts to zero-mean shocks without this adjustment would violate the stated zero-return assumption.

Residual assumption	P1 VaR (\$)	P2 VaR (\$)
Full residual covariance	\$50,286	\$15,747
Independent residuals	\$44,364	\$28,254

These are 5% order-statistic VaRs of simulated dollar P&L, not normal quantiles computed from the simulated sample standard deviations.

(d) Delta-normal comparison. From the stock return sample directly, the covariance matrix is:

	A	B
A	0.000285464	0.000211689
B	0.000211689	0.000230576

For $v_1 = 10^6(1, 1)^T$ and $v_2 = 10^6(1, -1)^T$, $\text{VaR}_{.05} = z_{.95} \sqrt{v^T \hat{\Sigma}_{AB} v}$.

Portfolio	Full simulation (\$)	Delta normal (\$)	Difference (\$)
P1	\$50,286	\$50,415	-\$128
P2	\$15,747	\$15,834	-\$87

Reconcile

(e) Direction and size of the shortcut's effect. Both directions agree with my prediction. Relative to the full simulation, independent residuals change P1 VaR by -\$5,923 (-11.78%) and P2 VaR by \$12,507 (79.42%). P2 is more exposed in both dollars and percentage terms for this sample.

The fitted market exposures are proportional to $\beta_A + \beta_B = 1.970695$ in P1 and $\beta_A - \beta_B = 0.099861$ in P2. Most of the common market factor cancels in P2. The following variance decomposition shows why residual dependence then matters so much; units are million dollars squared, i.e. each dollar-variance term is divided by 10^{12} .

Portfolio	Market term	Residual diagonal term	Residual covariance term
P1	0.00043248	0.00029924	0.00020769
P2	0.00000111	0.00029924	-0.00020769

The same covariance amount is added to P1 and subtracted from P2. Deleting it is a much larger proportional disturbance to the spread's small remaining variance. Separate regressions do not justify treating their residuals as independent: OLS orthogonality applies to each regression's residual and its regressor, not to the two residual series.

(f) Why simulation and delta-normal agree, and when they would not. With an intercept and a common sample, OLS makes the residuals orthogonal to the market. Using a consistent divisor gives the exact identity

$$\hat{\Sigma}_{AB} = \hat{\beta} \hat{\beta}^T s_{MKT}^2 + \hat{\Sigma}_{\epsilon}.$$

The largest absolute entrywise discrepancy when I reconstruct this identity is $1.626303\text{e-}19$. Under joint normality, orthogonality of the market and residual block is also independence. A linear combination of these normal inputs is normal with exactly the stock covariance above, so the full residual simulation and delta-normal method describe the same population P&L distribution.

The simulated VaRs differ from the analytic values by -0.255% for P1 and -0.548% for P2. These small differences arise from finite Monte Carlo sampling of a quantile, not a different fitted covariance. As another check, the independent-residual model's analytic VaRs are \$44,494 and \$28,507, also close to its simulated values.

Agreement would fail if the residual covariance were deleted, if market and residual inputs had material nonnormal joint tails, if the portfolio payoff or factor relationship were nonlinear, or if different samples, variance divisors, or means were used. With nonnormal inputs, an OLS covariance identity can still hold, but matching covariance no longer identifies the entire P&L distribution or its quantiles. OLS alone does not turn uncorrelated quantities into independent ones; the joint-normal assumption supplies that step.